

A Unified Framework for Robust Encrypted Malicious Traffic Detection in Adverse Environments via Graph Structure Learning

Jianjin Zhao¹, Zhiwei Cui¹, Junsong Fu¹, Meng Shen², *Member, IEEE*, and Qi Li¹

Abstract—The widespread adoption of encryption protocols enables attackers to conceal malicious activities within encrypted traffic, rendering traditional detection methods ineffective. Graph Neural Networks (GNNs) have emerged as a promising solution by modeling network objects and their interactions within graph representations to capture the collaborative behavioral patterns of complex threat activities. However, the well-performed premise of GNNs does not always hold in adverse environments, leading to unsatisfactory performance, suffering from three critical issues including (1) incomplete information analysis, where heterogeneous relations among network objects are often overlooked (2) lack of solutions for evasion techniques, as existing methods focus on robust representation learning but fail to correct adversarial distortions, and (3) limited robustness evaluation, relying on synthetic feature perturbations rather than raw traffic manipulations in line with real-world attacks. To address these issues, we propose RETA, a unified framework for robust encrypted malicious traffic detection via graph structure learning. First, RETA unifies heterogeneous subgraphs capturing semantic metapaths and homogeneous subgraphs modeling behavioral similarities among encrypted sessions and takes a tailored Heterogeneous Graph Attention Network (HAN) encoder for neighborhood information aggregation. Then, it employs a unified graph structure learning framework to correct noisy relations induced by evasion techniques through channel attention-based aggregation and Bayesian inference-based estimation. Following an iterative manner, RETA mutually improves relation modeling and detection robustness. Finally, RETA simulates various realistic adverse conditions by modifying raw traffic captures, ensuring comprehensive robustness evaluations against network fluctuations and adversarial attacks. Extensive experiments demonstrate the superior robustness of RETA, significantly improving detection performance in adverse environments. Even under extreme adverse conditions (i.e., 30% packet loss rate and 5 perturbation edges), RETA still shows significant advantages, delivering 8.94% and 4.85% accuracy improvements over the baseline models on average.

Index Terms—Encrypted traffic, malicious traffic detection, graph structure learning.

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I. INTRODUCTION

WITH the rapid proliferation of smart devices and internet applications, global network traffic has experienced explosive growth. To ensure secure data transmission, encrypted traffic becomes the mainstream form of network communications. However, the widespread deployment of encryption techniques also expands the attack surface, as attackers increasingly conceal threat activities (e.g., malicious payload transmission, covert channel construction, and sensitive data theft) within encrypted traffic. According to [1], between October 2023 and September 2024, the Zscaler cloud blocked 32.1 billion attacks embedded in SSL/TLS-encrypted traffic, accounting to 87.2% of all threats.

The ubiquitous adoption of encryption protocols presents significant challenges to cyberspace security. Traditional detection methods based on deep packet inspection and signature/rule matching are gradually becoming obsolete due to their inability to inspect encrypted payloads [2]. Consequently, research has shifted toward advanced machine learning and deep learning algorithms, leveraging feature engineering and representation learning techniques to extract discriminative patterns for threat detection. Recent advancements have primarily focused on exploring correlation analysis to capture the collaborative behavioral patterns of complex threat activities.

Graph Neural Networks (GNNs) [3], [4], [5], as an emerging representation learning technique, have achieved considerable progresses in encrypted malicious traffic detection by modeling fine-grained network objects and integrating complicated relations into graph representations [6], [7]. Various graph construction paradigms at packet/session/host levels have been proposed to model complex behavioral patterns for distinct tasks. Specifically, Shen et al. [8] represent packet-level interactions between clients and servers as traffic interaction graphs for decentralized application identification. Zhao et al. [9] construct encrypted traffic graphs to model service and behavior correlations among encrypted sessions for malicious flow detection. Lo et al. [10] formulate endpoint communications between hosts as network graphs for intrusion detection. Existing studies demonstrate that topological patterns extracted by GNNs are highly effective for threat detection.

However, prior studies [11], [12] have revealed that the strong expressive power of GNNs heavily relies on the premise that the observed graph topology is reliable and consistent with

the homophily property (i.e., similar nodes tend to form connections). In practice, this assumption is frequently violated due to observational errors and adversarial distortions. Regular network fluctuations can lead to packet loss, retransmission, and out-of-order delivery, distorting observable features (e.g., packet lengths). Additionally, various evasion techniques (e.g., adopting legitimate certificates) further obfuscate the correlations between network objects, undermining the integrity of graph topology. To adapt to the adversarial threat landscape, it is imperative to exploit the full potential of GNNs on noisy graphs for robust encrypted malicious traffic detection in adverse environments [13], [14].

Several studies have explored reliable graph construction for robust traffic analysis. Pujol-Perich et al. [15] model the inter-dependencies between flows traversing the network as graphs to capture underlying structural patterns against potential adversarial attacks. Li et al. [7], [16] adopt graph matching and propagation for robust encrypted traffic classification among various network environments in an iterative manner. Fu et al. [17], [18] construct a compact flow interaction graph to detect abnormal interaction patterns against typical evasion techniques. Zhao et al. [19] construct a relational multigraph to realize robust malicious flow detection via discriminative relation incorporation and misleading relation correction. Despite their effectiveness, three critical deficiencies persist in current graph-based approaches when applied to adverse environments.

- *Incomplete Information Analysis:* Existing methods commonly neglect the rich interactions among various network objects (e.g., fingerprints and certificates) of encrypted sessions, which reflect the reliability of web services to a certain extent.
- *Lack of Solutions for Evasion Techniques:* Evasion techniques obscure both features and interactions, while existing studies focus solely on deriving robust representations rather than correcting noisy relations, leading to unsatisfactory performance in adverse environments.
- *Limited Robustness Evaluation:* Few methods systematically simulate real-world adversarial attacks for robustness assessment. Simply modifying feature vectors rather than raw traffic captures is not in line with actual attacks, limiting their practical deployment.

To address these limitations, we propose RETA, a unified graph structure learning framework for robust encrypted malicious traffic detection in adverse environments. Specifically, RETA first constructs a comprehensive graph comprising heterogeneous and homogeneous subgraphs to model semantic metapaths and behavioral similarities among encrypted sessions. Then it takes a tailored Heterogeneous Graph Attention Network (HAN) encoder that to uniformly aggregate metapath-based and similarity-based neighborhood information, obtaining contextual session representations. To mitigate the impact of noisy interactions induced by evasion techniques, it retrieves session embeddings from intermediate layers to generate observations as guidance. RETA leverages channel attention-based aggregation and Bayesian inference-based estimation to reconcile conflicted observations, deriving an optimized graph with relation correction and completion. By iteratively refining the graph structure

along the HAN encoder, RETA simultaneously improves relation modeling and detection robustness.

For comprehensive robustness evaluation, we simulate various adverse conditions by directly modifying raw traffic captures, which is more inline with real attacks. Specifically, we randomly drop packets to obfuscate session features, simulating network fluctuations, and purposefully modify the data fields to obfuscate graph topology, simulating evasion techniques. Extensive experiments validates the superior robustness of RETA.

The key contributions of this paper can be summarized as follows:

- We construct a comprehensive graph integrating heterogeneous (metapath-based) and homogeneous similarity-based subgraphs to model the relations among encrypted sessions, and develop a tailored HAN to uniformly aggregate neighborhood information from both subgraphs.
- We propose a unified framework for robust encrypted malicious traffic detection. Following an iterative manner, both refined subgraph structures are derived along with the optimized HAN encoder, enabling mutual optimizations of relation modeling and detection robustness.
- We design practical evasion techniques by modifying raw traffic captures to simulate real-world attacks. Comprehensive evaluations demonstrate the superior robustness of RETA against network fluctuations and adversarial perturbations.

The remainder of this paper is organized as follows. We review the related studies in Section II, and introduce some preliminaries in Section III. Section IV presents the design of RETA. In Section V, the performance evaluations of RETA are provided. In Section VI, the limitations and potential improvements of RETA are discussed. Finally, we conclude in Section VII.

II. RELATED WORK

In this section, we review and summarize the relevant studies in the following three aspects: encrypted malicious traffic detection, graph neural network, and graph structure learning.

A. Encrypted Malicious Traffic Detection

The core challenges in encrypted malicious traffic detection arise from fundamental constraints in representing encrypted traffic. The encryption of payloads obscures semantic information, while the sparsity of available metadata results in high-dimensional feature distributions. These limitations hinder traditional detection methods, which not only suffer from combinatorial explosion but also struggle with capturing informative semantic association. Consequently, research in this area has shifted from experience-driven Machine Learning (ML)-based methods to data-driven Deep Learning (DL)-based paradigms. We first review ML-based and DL-based methods, followed by an overview of recent GNN-based advancements leveraging graph modeling to capture collaborative behavioral patterns for robust detection.

ML-based: Early research primarily relies on expert knowledge, employing manually designed feature sets combined with

machine learning algorithms for detection. The technical solutions at this stage focus on extracting network flow metadata, constructing a basic feature space by parsing packet feature sequences (e.g., lengths, arrival times), IP/TCP header fields (e.g., five-tuple, TTL values), and TLS handshake information (e.g., version, cipher suites). To further enhance representation capabilities, researchers introduced analytical metrics such as mean, variance, and quantiles to generate statistical features like packet length distribution and time interval variance, thereby achieving quantitative descriptions of traffic behaviors. Anderson et al. [20] conduct pioneering work in this field, proposing a supervised ML-based method that integrates TLS handshake metadata with DNS and HTTP contextual information, successfully enabling efficient detection of TLS-encrypted malware traffic. Subsequent works [21], [22], [23] refine flow-level feature design, while Fu et al. [24], [25] introduce frequency-domain representations to enhance feature stability. More recent advances explore richer structures: Yu et al. [26] employ attributed graphs with graph kernels for robust detection under data imbalance; Fu et al. [27] analyze packet length semantics to detect tunneled flooding attacks. Despite notable progress, ML-based methods remain highly dependent on manual feature design and show significant performance degradation under dynamic and evolving network environments.

DL-based: To mitigate manual feature reliance, DL-based methods map raw traffic into images, sequences, or graphs, enabling automated representation learning. Wang et al. [28] first apply representation learning to raw traffic data, classifying encrypted traffic by mapping it into images. In sequence modeling, recurrent neural networks (RNNs), Transformers, and other sequence models are widely used to capture long-range dependencies in packet feature sequences. Djaidja et al. [29] introduce an early intrusion detection method combining attention mechanisms and RNNs, achieving efficient early classification of encrypted malicious flows before session termination. In graph structure modeling, Shen et al. [8] proposed a decentralized application identification method, abstracting packet length sequences into graph structures for classification, further expanding the forms of encrypted traffic representation. These methods effectively reduce reliance on manual feature design and minimize information loss. Yang et al. [30] develop TrafCL, a contrastive learning framework leveraging session augmentation and co-attention mechanisms to robustly detect incomplete or obfuscated malicious encrypted traffic. Fu et al. [31] design pVoxel, an unsupervised method based on point cloud analysis that significantly reduces false positives of ML-based detection models by capturing the topological structure of traffic feature space. Zhang et al. [32] propose AOC-IDS, an online adaptive IDS using autoencoders with a contrastive learning loss and a labor-free continual adaptation framework, achieving strong adaptability to dynamic network environments. Alotaibi et al. [33] propose Mateen, an adaptive ensemble learning framework that enhances the adaptability of deep autoencoders to evolving benign network behaviors, achieving large gains in practical dynamic environments. Zhao et al. [34] design Trident, a universal framework for unknown and fine-grained encrypted traffic detection, using one-class learning and an ensemble of

modules to enable incremental updates and support fine-grained unknown class detection. Shen et al. [35] propose SmartDetector, which introduces a Semantic Attribute Matrix (SAM) representation combined with contrastive learning and data augmentation, improving robustness against obfuscation and new attack variants. However, Sun et al. [36] propose AdvTG, an adversarial traffic generation framework that combines large language models and reinforcement learning to generate semantically correct and functional adversarial traffic, effectively deceiving various DL-based detection models. When faced with complex challenges such as protocol version iterations and dynamic adversarial attacks in real-world network environments, existing methods still exhibit significant bottlenecks in representation robustness.

GNN-based: The recent efforts in encrypted malicious traffic detection primarily focus on exploring correlation analysis to capture collaborative behavioral patterns to improve robustness. Cui et al. [37] achieve high-precision association between IPv6 addresses and users using non-encrypted TLS handshake information based on Siamese heterogeneous graph attention networks. Pham et al. [6] design a method based on communication graph construction and deep GNNs, addressing issues such as dynamic changes in encrypted traffic and shared library interference by extracting packet header information. Fu et al. [38] introduce ST-Graph, which jointly exploits spatial, temporal, and graph structural features of encrypted traffic, achieving high precision and robustness in encrypted malware traffic detection across real-world deployments. Qiu et al. [39] propose a two-step feature decoupling and dynamic graph diffusion scheme, effectively resolving the entanglement of traffic feature distributions in network intrusion detection. Ding et al. [40] design a meta-learning graph attention network-based IoT intrusion detection system, optimizing graph partitioning strategies in large-scale IoT networks through offline reinforcement learning algorithms to achieve efficient parallel detection performance. Additionally, Fu et al. [17], [18] design an encrypted malicious traffic detection system based on flow interaction graph structure analysis, identifying unknown threats without labeled data. Mao et al. [41] propose a label-aware federated graph contrastive learning framework, significantly improving network intrusion detection performance in few-shot scenarios by introducing linear graph processing for flow embeddings combined with federated learning. Wu et al. [42] propose TCG-IDS, a self-supervised IDS framework based on temporal contrastive graph learning, achieving superior performance and low false positive rates under multi-class network intrusion detection scenarios. Zhang et al. [43] design MH-Net, which constructs multi-view heterogeneous traffic graphs and employs contrastive learning to enhance encrypted traffic classification under complex correlations across traffic bytes. Cui et al. [44] propose FG-SAT, which builds flow graphs and uses GraphSAGE and GAT for encrypted traffic classification under environment shifts, and employs feature selection based on Jensen-Shannon divergence to improve robustness. However, it is important to emphasize that existing methods have yet to fully account for data noise interference caused by network fluctuations and evasion techniques.

TABLE I
COMPARISON WITH ADVANCED RELATED WORK

Categories	Ref.	CA	AS	NM	RE	UO
ML-based	WHISPER [24], [25]	×	✓	×	✓	×
	TLS-F [26]	×	×	×	✓	×
	EXOSPHERE [27]	×	✓	×	✓	×
DL-based	AOC-IDS [32]	×	×	×	✓	×
	EARLY [29]	×	×	×	×	×
	SMARTDETECTOR [35]	✓	✓	×	×	×
GNN-based	TCG-IDS [42]	✓	×	×	×	×
	MH-NET [43]	✓	×	×	×	✓
	FG-SAT [44]	✓	✓	×	✓	×
	RETRIAL [19]	✓	✓	✓	✓	×
	RETA	✓	✓	✓	✓	✓

CA: Correlation Analysis; AS: Attack Simulation; NM: Noise Mitigation;
RE: Robustness Evaluation; UO: Unified Optimization;

Unfortunately, existing GNN-based methods predominantly rely on an ideal assumption of data integrity, failing to account for real-world adversarial scenarios where both node features and graph structures can be compromised. In practical cyberspace full of uncertainties and adversarial threats, encrypted sessions often exhibit incomplete feature representations and spurious inter-dependencies due to intentional obfuscation techniques and protocol manipulations. Considering data quality as the cornerstone of AI-driven security systems, we propose a unified graph structure learning framework for robust encrypted malicious traffic detection, namely RETA. It iteratively optimizes both homogeneous and heterogeneous relations and the detection model to mutually improve relation modeling and detection robustness. Additionally, we simulate various adverse conditions in line with actual attacks by directly modifying raw traffic to rigorously validate RETA's robustness. Table I compares RETA with some advanced related work. Though many state-of-the-art methods have made notable progress in correlation analysis (CA) and robustness evaluation (RE), they often overlook key aspects such as noise mitigation (NM), attack simulation (AS), and unified optimization (UO). In contrast, RETA uniquely offers an end-to-end robust detection pipeline by jointly optimizing graph structure and detection model, achieving comprehensive capabilities across all evaluated dimensions. This highlights its superior adaptability and resilience in real-world adversarial and noisy network environments.

B. Graph Neural Networks

To tackle non-Euclidean data with more expressive power, GNNs are proposed to learn the complex relations and the rich inter-dependencies among entities in the graph domain. Due to their strong performance and interpretability, GNNs have recently been widely applied in various fields (e.g., social networks [45], recommendation systems [46] and security applications [47]). Among foundational GNN architectures, Graph Convolutional Network (GCN) [48] bases on first-order approximations of spectral graph convolutions to aggregate neighborhood information and enlarge the receptive field for each node, achieving promising performance in many tasks. However, GCN has only been applied in transductive setting with fixed graphs. To fix this issue, Hamilton et al. [49] propose

GraphSAGE that generalizes GCN to inductive learning. Instead of simple convolutions, aggregator functions are trained to generate embeddings from nodes' local neighborhoods. GraphSAGE can effectively trade off performance and runtime by sampling node neighborhoods. Velickovic et al. [50] introduce an attention mechanism to GNNs and proposed Graph Attention Network (GAT), which can implicitly specify different weights to different neighbors without knowing the graph structure. For graph-level classification, Li et al. [51] incorporate hierarchical transitive alignment of nodes and transforms variable-sized graphs into fixed-size aligned density matrices to encode both topological structures and quantum interference patterns into graph representations. Generally, most GNNs depend on the homophily assumption that nodes belonging to the same class are more likely to be connected. GNNs for heterophilic graphs are gaining increasing research attention [52] to enhance graph learning with heterophily. Li et al. [53] introduce Haar-type graph framelets with permutation equivariance, sparsity, and efficient computation for heterophilous graphs.

Moreover, heterogeneous graphs, also called Heterogeneous Information networks (HINs) with multi-typed nodes and edges, contain more abundant information and are more in line with the real-world scenario. Wang et al. [54] propose Heterogeneous Graph Attention Network (HAN) to perform a hierarchical attention mechanism respectively on node level and semantic level to aggregate metapath-based neighbors and obtain the final node embeddings. To consider the intermediate nodes in metapaths, Fu et al. [55] propose Metapath Aggregated Graph Neural Network (MAGNN). It first transforms heterogeneous node embeddings into the same dimension, then applies intra-metapath and inner-metapath aggregations with an attention mechanism to learn the node embeddings.

This work focuses on adapting advanced graph representation learning techniques to security applications rather than developing new GNN architectures. Specifically, we propose a general HGNN-powered framework for encrypted malicious traffic detection and take HAN as an example to present its implementation. Notably, our framework supports seamless substitution of alternative state-of-the-art HGNN models for node classification tasks, preserving methodological generality while maintaining detection performance. This approach positions heterogeneous graph modeling as a flexible solution adaptable to evolving threat detection requirements.

C. Graph Structure Learning

Graph Structure Learning (GSL) [56] aims to optimize the incomplete and noisy graph structures for better performance of GNN on downstream tasks. Chen et al. [57] first propose an Iterative Deep Graph Learning (IDGL) framework to optimize graph structures. Specifically, IDGL utilizes cosine similarity as the edge weights and controls the sparsification of the generated graph with a preset threshold. Wang et al. [58] propose Graph structure Estimation neural Network (GEN). GEN firstly hypothesizes that the optimal graph structure conforms to Stochastic Block Model (SBM) [59]. Then GEN generates a set of multi-view observations as guidance to compute the posterior

probability of the optimal graph based on Bayesian inference. Finally, GEN optimizes the parameters based on Expectation-Maximization (EM) algorithm, achieving mutual optimizations of both graph structure and GNN parameters. Zhao et al. [60] extend GSL to HGNN and propose a Heterogeneous Graph Structure Learning framework, namely HGSL. Based on IDGL, HGSL introduces feature propagation graphs and utilizes meta-path2vec [61] to generate semantic graphs as supplementation. In each heterogeneous relation, feature similarity graphs, feature propagation graphs, and semantic graphs are aggregated by channel attention operation and fused to generate the optimal heterogeneous graph. Motivated by these studies, we introduce graph structure learning to mitigate the perturbations introduced by potential evasion techniques in adverse environments, mutually improve relation modeling and detection robustness for encrypted malicious traffic detection.

III. PRELIMINARIES

In this section, we first define the threat model by presenting the adversaries' capabilities and corresponding attack scenario simulations, and then introduce the background knowledge about SSL/TLS handshake process related to our work.

A. Threat Model

In this work, we consider two types of adversaries: *passive* and *proactive*. The *passive* adversary prioritizes disruption over evading detection, causing packet loss, retransmission, and out-of-order delivery, like general network fluctuations. The loss and distortion of available information affect detection performance. The *proactive* adversary actively manipulates malicious traffic to evade detection by exploiting vulnerabilities inherent in learning-based models.

To systematically evaluate the impact of these adversaries, we consider two corresponding attack scenarios: (1) packet loss due to network fluctuations or Denial of Services (DoS) attacks and (2) graph structure perturbations via adversarial attacks. In the former scenario, we randomly drop packets at a certain rate, simulating the *passive* adversary. As for the latter scenario, we add perturbation relations to modify the graph structure, simulating the *proactive* adversary.

B. TLS Handshake

TLS protocol is widely employed to encrypt the communications between web applications and servers. Previous studies primarily leverage the parameters in the unencrypted TLS handshake messages negotiated for subsequent encrypted communications. The details of the informative TLS handshake messages are as follows:

ClientHello message: Before the encrypted communications, the client sends a ClientHello message to the server to initiate a TLS handshake, which contains the maximum version of SSL/TLS protocol that the client supports, a 32-bytes ClientHello random byte string, the list of cipher suites supported, the list of compression algorithms supported and some optional extension.

ServerHello message is a reply to the ClientHello message. It contains the version of SSL/TLS protocol that the server chooses, a 32-bytes ServerHello random byte string, the lists of cipher suites and compression algorithms chosen from the list provided by the client, and the extensions in response.

Certificate message is sent by the server following the ServerHello message, which contains a chain of certificates beginning with the sender's certificate and ending with the Root certificate. Each certificate in the chain is signed by the one preceding it. The client verifies the server's identity based on the certificate chain. It contains much valuable information like the subject, issuer information, and the validity of certificates.

Previous studies [20], [62], [63] have demonstrated that unencrypted TLS handshake messages can be utilized to identify threats. Specifically, the distributions of cipher suites and extensions differ between encrypted malicious and benign traffic. Certain cipher suites and extensions are considered insecure and rarely used by enterprises. Additionally, the status of a certificate, whether self-signed or expired, reflects the reliability of web services. Domain names within Server Name Indication (SNI) extensions or certificates can also be extracted for malicious domain detection. However, with the development of evasion techniques, these detection approaches progressively become invalidate.

IV. METHODOLOGY

A. Framework Overview

In this section, we present the high-level design of RETA. Generally, correlation analysis effectively mitigates the feature noise introduced by network fluctuations, while various evasion techniques further obfuscate the relations among network objects, leading to unsatisfactory detection performance. Thus, in RETA, we construct a graph to comprehensively model the relations among encrypted sessions, and adaptively correct those noisy ones via graph structure learning in a unified way. Following an iterative manner, it realizes mutual optimization with noisy relation correction and detection robustness improvement. Fig. 1 shows four key components of RETA, i.e., graph construction, observation generation, graph derivation, and iterative optimization.

Graph Construction: RETA collects and splits raw traffic captures into encrypted sessions for graph construction. Meanwhile, it parses unencrypted TLS handshake messages and per-packet features to characterize both client-side and server-side configurations and network behaviors. In the graph, it utilizes network objects (i.e., sessions, fingerprints and certificates) as nodes, and establishes connections by modeling semantic and behavioral relations among sessions. The graph can be further subdivided into homogeneous and heterogeneous subgraphs by relations. More details about graph construction are presented in Section IV-B.

Observation Generation: RETA adopts HAN to aggregate spatial contextual information within the graph. To infer a better graph topology with relation correction and completion, we retrieve session embeddings from intermediate layers to construct multi-order neighborhood similarity graphs as observations.

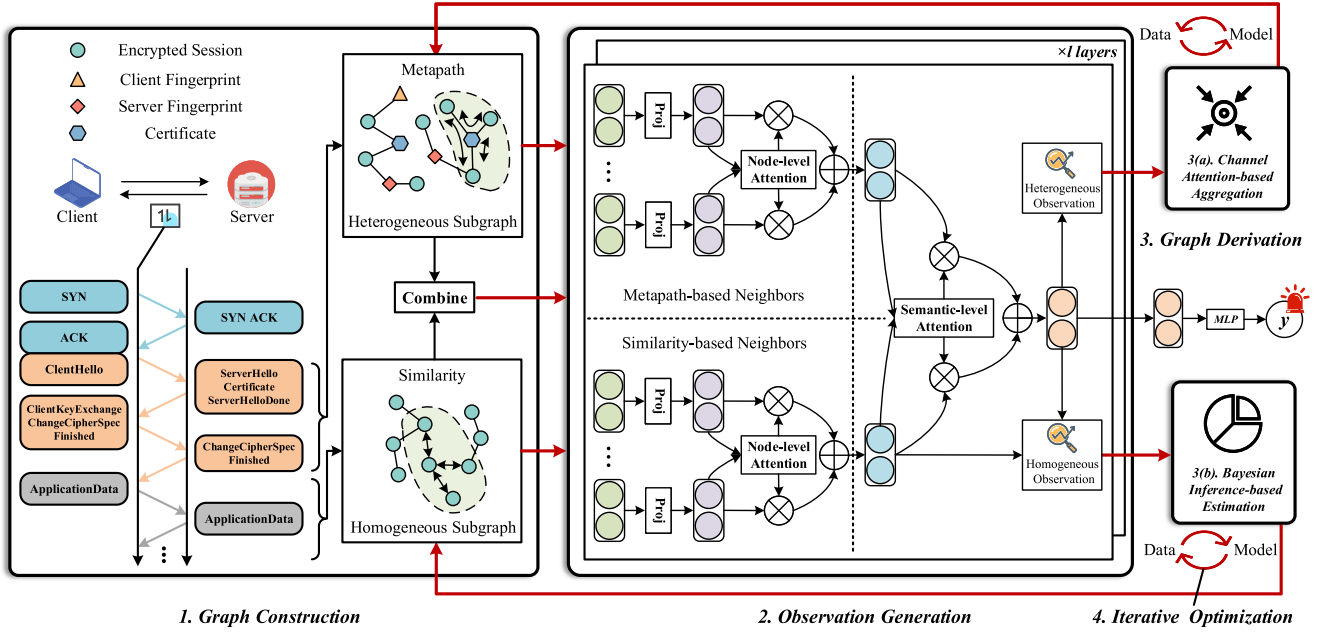


Fig. 1. The overall framework of RETA.

Following the hierarchical attention mechanism in HAN, both node-level and semantic-level session embeddings in different layers are collected to generate observations and jointly guide the optimal graph derivation. We will describe the details of the observation generation in HAN in Section IV-C.

Graph Derivation: In RETA, different graph derivation strategies are employed based on different relations. Specifically, channel attention-based aggregation and Bayesian inference-based estimation are applied on heterogeneous and homogeneous observations respectively to derive better graph topology, replacing the original graph for further optimization of the graph encoder. We will provide the details of graph derivation of both heterogeneous and homogeneous observations in Section IV-D.

Iterative Optimization: With the temporary optimal graph, we can further optimize the HAN encoder, retrieve better node embeddings from hidden layers, and thus generate more accurate observations. Then a new iteration begins until convergence. Following an iterative manner, RETA not only generates optimal graph topology with relation correction and completion, but also enhance the representation ability of HAN in encrypted malicious traffic detection with detection robustness improvement. We will detail how RETA iteratively optimizes the data and model in Section IV-E.

B. Graph Construction

To model the rich relations among encrypted sessions, we construct a graph incorporating various network objects as nodes, and both semantic and behavioral relations as edges.

1) **Nodes:** We extract four types of nodes to denote the network objects in TLS communications, including encrypted session node S , client fingerprint node F_c , server fingerprint node F_s , certificate node C . The corresponding node features are as follows:

Encrypted Session Node S denotes an encrypted session, including both uplink and downlink packets that share the same 5-tuple (source IP, source port, destination IP, destination port, protocol). The features include the parameters negotiated during TLS handshake phase for subsequent encrypted communication and the statistical data of uplink and downlink packets after the session is established.

Client Fingerprint Node F_c reflects the configuration of the client application initiating TLS sessions. The features include supported version, accepted cipher suites, list of extensions, elliptic curves, and elliptic curve formats in ClientHello message, with the MD5 hash value forming a 32-digit hexadecimal JA3 fingerprint for correlations.

Server Fingerprint Node F_s reflects the security configurations of the accessed server. The features include selected version, cipher suites, and extensions in ServerHello message, with the MD5 hash value forming a 32-digit hexadecimal JA3s fingerprint for correlations.

Certificate Node C denotes the certificate issued by a Certificate Authority (CA) to verify the server's identity. The features include the issuer (i.e., non-root issuing CA), the subject (i.e., the entity of the certificate owner), the validity period, and the number of certificates in the Certificate message.

2) **Relations:** Generally, threat activities like Distributed Denial of Service (DDoS) and botnet attacks involve multiple encrypted sessions collaboratively generated by a collection of devices. Therefore, analyzing each encrypted session in isolation for threat detection is inadequate. It is essential to consider the complex relations among encrypted sessions from a global perspective. The graph can be subdivided into heterogeneous and homogeneous subgraphs by relations. Based on the TLS protocol, we propose four metapath-based correlations (**C1-C4**) with rich semantics to construct heterogeneous subgraphs. Besides, we introduce DTW and Jaccard distances to measure the

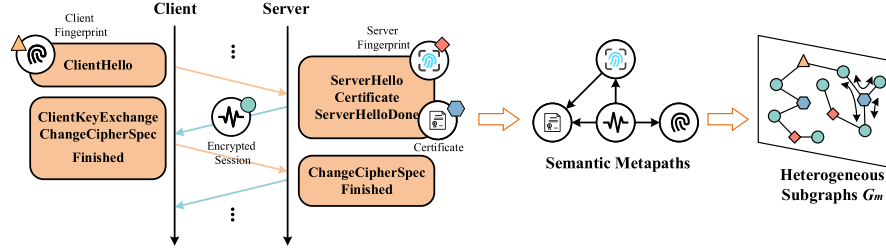


Fig. 2. Heterogeneous subgraph construction.

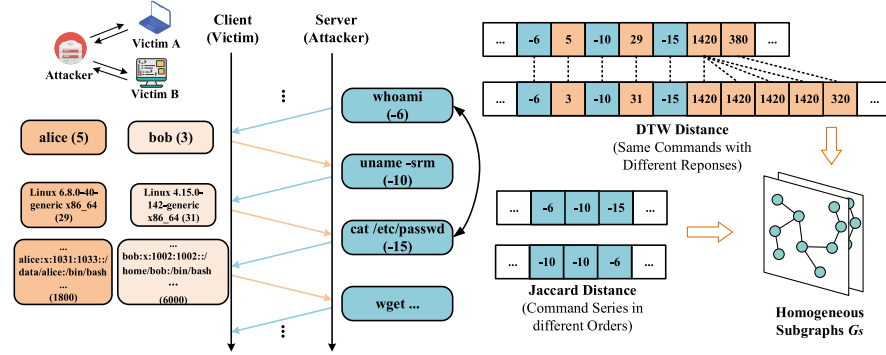


Fig. 3. Homogeneous subgraph construction.

similarity-based correlations (**C5-C6**) in network behaviors and construct homogeneous subgraphs.

Specifically, during the TLS handshake process, multiple unencrypted fields including versions, cipher suites negotiate many unencrypted fields including version, cipher suites and certificate chains, are negotiated as critical indicators of web service reliability. We systematically leverage these observable plaintext parameters to construct heterogeneous subgraphs that encode multi-relational behavioral patterns. Fig. 2 illustrates the construction of heterogeneous subgraphs based on semantic metapaths defined as follows:

- C1: SF_cS metapath** denotes that two encrypted sessions share the same JA3 fingerprint, which represents that they are initiated by similar client applications like web browsers, malware samples, or basic Software Development Kits (SDKs).
- C2: SF_sS metapath** denotes that two encrypted sessions share the same JA3s fingerprint, which represents that the server-side configurations are similar.
- C3: SCS metapath** denotes that two encrypted sessions share the same certificate, which represents that the identities of the web services accessed are the same and issued by the same non-root issuer.
- C4: SF_sCF_sS metapath** denotes that the server fingerprints of two encrypted sessions share the same certificate, which represents that the web services sharing similar server configurations have the same certificate.

In addition, bidirectional packet length sequences are widely applied to characterize network behaviors [8], [64]. For example, bursts of uplink packets may indicate activities like google drive uploads or data exfiltration, while bursts of downlink packets may be associated with web browsing or malware delivery. We

utilize DTW and Jaccard distance metrics to quantify the behavioral similarities among encrypted sessions. The behavioral similarities within these homogeneous subgraphs are defined as follows:

C5: DTW distance similarity. Dynamic Time Warping algorithm quantifies the similarity between temporal sequences by aligning them non-linearly. Given the stability of downlink packets and the variability of uplink packets, we construct a k -NN graph of encrypted sessions based on DTW distance of bidirectional packet length sequences.

C6: Jaccard distance similarity. Jaccard distance measures the dissimilarity between sample sets by computing the ratio of the difference between their union and intersection to the union size. Given the unordered nature of C&C instructions, we construct a k -NN graph of encrypted sessions by computing the Jaccard distance of downlink packet length sequences.

Threat activities typically adhere to fixed procedures to establish connections, leading to relatively consistent communication patterns. Inspired by [65], we extend the application scenario in user action identification to threat identification. To capture these temporal correlations, we employ DTW distance to measure the behavioral similarity of bidirectional encrypted communication patterns. Additionally, since C&C instructions are often executed in non-strict sequence, we introduce Jaccard distance to measure the similarity of downlink packet length sequences, capturing spatial correlations between encrypted sessions.

Fig. 3 illustrates the construction of homogeneous subgraphs based on DTW and Jaccard distance metrics. Different victims may respond differently to the same command due to the variations in operation system versions, host configurations, and

network conditions. For example, the ‘whoami’ command may return ‘alice’ on one victim and ‘bob’ on another. Notably, due to the constraints imposed by the Maximum Transmission Unit (MTU), TLS messages frequently span multiple TCP segments, making DTW distance particularly effective in capturing temporal dependencies. Meanwhile, the attacker can arbitrarily reorder command series. Unlike sequential distance metrics, Jaccard distance provides a robust measure of set similarity that remains unaffected by order, enabling effective identification of such manipulations.

C. Observation Generation

Due to the observational errors (e.g., performance bottlenecks) and random errors (e.g., evasion techniques), the constructed graph is unreliable. To improve the graph structure and restore the true nature of the complex relations among network objects, we utilize the original graph along with multi-order neighborhood similarity graphs as multi-view observations of the optimal graph, guiding its derivation.

RETA takes HAN as the encoder. Specifically, we input the original graph $\mathbf{G} = (\mathbf{G}_m, \mathbf{G}_s)$ into HAN, which follows a hierarchical attention mechanism. At each layer, node-level attention first assigns importance scores to metapath-based neighbors, aggregating them into semantic-specific node representations for each metapath. Next, semantic-level attention assigns importance scores to each metapath, aggregating semantic-specific node representations into final comprehensive representations. In this context, we treat similarity-based neighbors in homogeneous graphs as metapath-based neighbors to align with the aggregation mechanism in HAN. The one-layer aggregations of node i in HAN are as follows:

$$h_i^\phi = \mathbf{N}\mathbf{A}_{j \in N_i^\phi}(h_j), \quad (1)$$

$$h'_i = \mathbf{S}\mathbf{A}_{\phi \in \Phi}(h_i^\phi), \quad (2)$$

where $\mathbf{N}\mathbf{A}$ and $\mathbf{S}\mathbf{A}$ denote node-level and semantic-level attention aggregations, N_i^ϕ denotes the collection of node i ’s metapath ϕ -based neighbors, and Φ denotes the collection of metapaths. During the hierarchical aggregations, the updated node representation captures both the structural information within its multi-order neighbors and the semantic information of various metapaths. We retrieve the node representations from intermediate layers to construct multi-order neighborhood similarity graphs as multi-view observations $\mathbf{O} = \{\mathbf{O}_m, \mathbf{O}_s\}$. Both heterogeneous subgraph $\mathbf{G}_m$ and homogeneous subgraph $\mathbf{G}_s$ can be further divided by metapaths and similarities.

For heterogeneous subgraph $\mathbf{G}_m$ divided by metapaths, cosine similarity is used to compute the similarity graphs of semantic-level features for each metapath $\phi \in \Phi$ to form heterogeneous observations $\mathbf{O}_m$. Firstly, we design a type-specific transformation to unify the heterogeneous nodes with different feature spaces as follows:

$$h_i^* = \sigma(\mathbf{W}_{t(v_i)}h'_i + \mathbf{b}_{t(v_i)}), \quad (3)$$

where $\sigma(\cdot)$ denotes a non-linear activation function, $t(\cdot)$ denotes a node type mapping function, $\mathbf{W}_{t(v_i)}$ and $\mathbf{b}_{t(v_i)}$ denote the

mapping matrix and the bias vector of node v_i ’s type. Then we apply K -head cosine similarity to generate metapath-specific feature similarity graphs as follows:

$$O_m^\phi[i, j] = \frac{1}{K} \sum_k \cos(\mathbf{w}_{k, \phi} \odot h_i^*, \mathbf{w}_{k, \phi} \odot h_j^*), \quad (4)$$

where $\odot$ denotes the Hardward product operation, $\mathbf{w}_{k, \phi}$ denotes a learnable weight matrix of metapath ϕ .

For homogeneous subgraph $\mathbf{G}_s$ divided by similarities, k -NN graphs are generated on node-level and semantic-level node embeddings to jointly form homogeneous observations $\mathbf{O}_s$. We directly take the homogeneous node embeddings to generate similarity-specific KNN graphs as follows:

$$\mathbf{O}_s^\phi = \{\mathbf{KNN}(\mathbf{h}^\phi), \mathbf{KNN}(\mathbf{h}')\}, \quad (5)$$

where $\mathbf{KNN}(\cdot)$ denotes a KNN graph generating function that takes feature vectors as input. It is notable that layer mark (l) is omitted for simplicity in the above presentation.

Considering a l -layer HAN, the heterogeneous observations $\mathbf{O}_m$ consists $|\Phi_m| \times (1 + l)$ heterogeneous feature similarity graphs, where $|\Phi_m|$ are original graph structures and $|\Phi_m| \times l$ are generated on semantic-level aggregated embeddings. The homogeneous observations $\mathbf{O}_k$ consists of $|\Phi_s| \times (1 + l) + l$ homogeneous feature similarity graphs, where $|\Phi_s|$ are original graph structures, $|\Phi_s| \times l$ are generated on node-level aggregated embeddings, and l are generated on semantic-level aggregated embeddings. Notably, these l homogeneous feature similarity graphs generated on semantic-level aggregated embeddings are shared by all homogeneous relations. The above observations serve as guidance to help correct the observed graph and infer a more trustworthy graph topology.

D. Graph Derivation

Though the multi-view observations reflect the optimal graph from different perspectives, there may exist conflicts among observations. The derivation of the optimal graph can be decoupled into independent subproblems focused on deriving its optimal components.

1) *Heterogeneous Subgraph Derivation*: Given heterogeneous observations $\mathbf{O}_m$, we aggregate them in each metapath $\phi \in \Phi_m$ independently to derive the estimated adjacency matrices $\mathcal{A}_m^\phi$ of the optimal heterogeneous subgraph $\mathcal{G}_m^\phi$ as follows:

$$\begin{aligned} \mathcal{A}_m^\phi &= \mathbf{AGG}(\mathbf{O}_m^\phi) \\ &= \mathbf{AGG}(\mathbf{A}_m^\phi, \mathbf{O}_m^{\phi, (1)}, \mathbf{O}_m^{\phi, (2)}, \dots, \mathbf{O}_m^{\phi, (l)}), \end{aligned} \quad (6)$$

where $\mathbf{AGG}$ denotes aggregation operation that aggregates both the original observed graph and the multi-order neighborhood similarity graphs in each metapath to derive the optimal heterogeneous subgraph.

2) *Homogeneous Subgraph Derivation*: Given homogeneous observations $\mathbf{O}_s$, inferring the optimal homogeneous subgraph $\mathcal{G}_s$ is challenging. However, assuming that $\mathcal{G}_s$ is known, it is simpler to calculate the probabilities of mapping it to the

observations. The inverse problem can be solved by Bayesian inference.

First, we suppose that the optimal homogeneous subgraph $\mathcal{G}_s$ satisfies the property of the Stochastic Block Model (SBM) with latent community structure to ensure homophily. Given SBM parameters Ω , known labels $\mathcal{Y}_L$, unknown predictions $\mathbf{Z}$, the probability distribution of the optimal homogeneous subgraph $\mathcal{G}_s$ is denoted as follows:

$$\begin{aligned} P(\mathcal{G}_s | \Omega, \mathcal{Y}_L, \mathbf{Z}) &= \prod_{\phi \in \Phi_s} P(\mathcal{G}_s^\phi | \Omega, \mathcal{Y}_L, \mathbf{Z}) \\ &= \prod_{\phi \in \Phi_s} \prod_{i \neq j} \Omega_{c_i c_j}^{\mathcal{G}_s^\phi[i,j]} (1 - \Omega_{c_i c_j})^{1 - \mathcal{G}_s^\phi[i,j]}, \end{aligned} \quad (7)$$

where

$$c_i = \begin{cases} y_i & \text{if } v_i \in \mathbf{V}_L, \\ z_i & \text{otherwise.} \end{cases} \quad (8)$$

Next, we introduce homogeneous observations $\mathbf{O}_s$ as external knowledge to infer the optimal homogeneous subgraph $\mathcal{G}_s$. Assuming that the edges in observations follows a Bernoulli distribution, its parameters are conditional on the existence of the edges in the optimal homogeneous subgraph, where α denotes the true positive rate, the probability that edges in the observations truly appear in $\mathcal{G}_s^\phi$, and β denotes the false positive rate. Supposing that the edge between node i and node j is observed N_{ij} times out of N (i.e., $|\mathbf{O}_s^\phi|$) observations, given the optimal homogeneous subgraph $\mathcal{G}_s^\phi$ and the parameters α, β , the probability of homogeneous observations can be calculated as follows:

$$\begin{aligned} P(\mathbf{O}_s^\phi | \mathcal{G}_s^\phi, \alpha, \beta) &= \prod_{i \neq j} [\alpha^{N_{ij}} (1 - \alpha)^{N - N_{ij}}]^{\mathcal{G}_s^\phi[i,j]} \\ &\quad \times [\beta^{N_{ij}} (1 - \beta)^{N - N_{ij}}]^{1 - \mathcal{G}_s^\phi[i,j]}. \end{aligned} \quad (9)$$

As such, we can utilize Bayesian inference to calculate the posterior probability of the optimal homogeneous subgraph $\mathcal{G}_s^\phi$ as follows:

$$\begin{aligned} P(\mathcal{G}_s^\phi, \Omega, \alpha, \beta | \mathbf{O}_s^\phi, \mathcal{Y}_L, \mathbf{Z}) &= \\ \frac{P(\mathbf{O}_s^\phi | \mathcal{G}_s^\phi, \alpha, \beta) P(\mathcal{G}_s^\phi | \Omega, \mathcal{Y}_L, \mathbf{Z}) P(\Omega) P(\alpha) P(\beta)}{P(\mathbf{O}_s^\phi, \mathcal{Y}_L, \mathbf{Z})}, \end{aligned} \quad (10)$$

where $P(\Omega)$, $P(\alpha)$, $P(\beta)$ and $P(\mathbf{O}_s^\phi, \mathcal{Y}_L, \mathbf{Z})$ are the probabilities of parameters, observations, labels, and predictions, which are assumed to be independent. To eliminate the effects of $\mathcal{G}_s^\phi$, we sum up all possible structures to derive the posterior probability of parameters Ω, α, β as follows:

$$P(\Omega, \alpha, \beta | \mathbf{O}_s^\phi, \mathcal{Y}_L, \mathbf{Z}) = \sum_{\mathcal{G}_s^\phi} P(\mathcal{G}_s^\phi, \Omega, \alpha, \beta | \mathbf{O}_s^\phi, \mathcal{Y}_L, \mathbf{Z}). \quad (11)$$

By maximizing the posterior probability, we derive the probability distribution $q(\mathcal{G}_s^\phi)$ to calculate the estimated adjacency matrices $\mathcal{A}_s^\phi$ as follows:

$$\mathcal{A}_s^\phi[i, j] = \sum_{\mathcal{G}_s^\phi} q(\mathcal{G}_s^\phi) \mathcal{G}_s^\phi[i, j] \quad (12)$$

To reduce computational overhead and potential noise, we mask (i.e., zero out) elements in the estimated adjacency matrices of both heterogeneous and homogeneous subgraphs that are smaller than the given threshold ϵ as follows:

$$\mathcal{A}[i, j] = \begin{cases} \mathcal{A}[i, j] & \text{if } \mathcal{A}[i, j] \geq \epsilon, \\ 0 & \text{otherwise.} \end{cases} \quad (13)$$

E. Iterative Optimization

The estimated adjacency matrices $\mathcal{A}$ and the graph encoder parameters Θ are optimized iteratively. Along with the graph encoder optimization, the node representations from intermediate layers are increasingly informative, providing more accurate observations and better estimated graphs. Conversely, the graph encoder can be further optimized with the updated graph. In this manner, RETA achieves mutual data-model optimizations, with noisy relation correction and detection robustness improvement.

1) *Update Θ* : In this work, we construct a graph to model the complex relations among encrypted sessions, where a small number of encrypted session nodes are labeled. RETA takes HAN as the graph encoder to classify encrypted sessions into malicious and benign. In semi-supervised node classification settings, the graph encoder is optimized by minimizing the cross entropy loss over all labeled nodes as follows:

$$\min_{\Theta} \mathcal{L}(\mathbf{A}, \mathbf{X}, \mathcal{Y}_L) = - \sum_{v_i \in \mathbf{V}_L} y_i \ln z_i, \quad (14)$$

where y_i and z_i are the label and prediction of node v_i . With the guide of all labeled nodes $v_i \in \mathbf{V}_L$, the graph encoder parameters Θ are updated via Stochastic Gradient Descent (SGD), allowing for learning more informative node representations.

2) *Update $\mathbf{A}$* : With the graph encoder parameters Θ held constant, we retrieve a new set of observations $\mathbf{O}$ as guidance to optimize the graph structure, where the optimal heterogeneous and homogeneous subgraphs are derived independently.

Given heterogeneous observations $\mathbf{O}_m$, channel attention is utilized to aggregate the multi-order neighborhood similarity graphs in each metapath $\phi \in \Phi_m$ to estimate the optimal adjacency matrices $\mathcal{A}_m$ for heterogeneous subgraph $\mathcal{G}_m$, where

$$\mathbf{O}_m^\phi = \{\mathbf{A}_m^\phi, \mathbf{O}_m^{\phi,(1)}, \mathbf{O}_m^{\phi,(2)}, \dots, \mathbf{O}_m^{\phi,(l)}\}. \quad (15)$$

The importance of multi-faced observations is balanced with channel attention to derive the optimal heterogeneous subgraph as follows:

$$\mathcal{A}_m^\phi = \mathbf{CA}(\mathbf{A}_m^\phi, \mathbf{O}_m^{\phi,(1)}, \mathbf{O}_m^{\phi,(2)}, \dots, \mathbf{O}_m^{\phi,(l)}), \quad (16)$$

where $\mathbf{CA}$ denotes channel attention operation to perform 1×1 convolution on the multi-view observations generated in different layers.

Given homogeneous observations $\mathbf{O}_s$, we apply the Expectation Maximization (EM) algorithm to maximize the posterior probability of the optimal homogeneous subgraphs $\mathcal{G}_s^\phi$ in (10), which is equivalent to maximize its logarithm form. In EM algorithm, E-step and M-step are repeated iteratively for Maximum Likelihood Estimation (MLE).

In E-Step, we construct a lower-bound in (17). Consider the hidden variable $\mathcal{G}_s^\phi$, the distribution of which is denoted by $q(\mathcal{G}_s^\phi)$ ($\sum_{\mathcal{G}_s^\phi} q(\mathcal{G}_s^\phi) = 1, q(\mathcal{G}_s^\phi) \geq 0$). Since logarithm is strictly concave, we can achieve the lower-bound according to Jensen's inequality as follows:

$$\begin{aligned} & \log P(\Omega, \alpha, \beta | \mathbf{O}_s^\phi, \mathcal{Y}_L, \mathbf{Z}) \\ &= \log \sum_{\mathcal{G}_s^\phi} q(\mathcal{G}_s^\phi) \frac{P(\mathcal{G}_s^\phi, \Omega, \alpha, \beta | \mathbf{O}_s^\phi, \mathcal{Y}_L, \mathbf{Z})}{q(\mathcal{G}_s^\phi)} \\ &\geq \sum_{\mathcal{G}_s^\phi} q(\mathcal{G}_s^\phi) \log \frac{P(\mathcal{G}_s^\phi, \Omega, \alpha, \beta | \mathbf{O}_s^\phi, \mathcal{Y}_L, \mathbf{Z})}{q(\mathcal{G}_s^\phi)}, \quad (17) \end{aligned}$$

where the equality holds when the random variable is constant. By summation, we get $q(\mathcal{G}_s^\phi)$ as follows:

$$q(\mathcal{G}_s^\phi) = \frac{P(\mathcal{G}_s^\phi, \Omega, \alpha, \beta | \mathbf{O}_s^\phi, \mathcal{Y}_L, \mathbf{Z})}{\sum_{\mathcal{G}_s^\phi} P(\mathcal{G}_s^\phi, \Omega, \alpha, \beta | \mathbf{O}_s^\phi, \mathcal{Y}_L, \mathbf{Z})}. \quad (18)$$

With the choice of $q(\mathcal{G}_s^\phi)$, Equation (17) provides the lower bound of the posterior probability. Next, we update the parameters to maximize the lower bound.

In M-Step, we hold $q(\mathcal{G}_s^\phi)$ constant and take the derivatives of the right side of (17) to have Maximum-A-Posteriori (MAP) estimations as follows:

$$\sum_{\mathcal{G}_s^\phi} q(\mathcal{G}_s^\phi) \sum_{i \neq j} \left[\frac{\mathcal{G}_s^\phi[i, j]}{\Omega_{c_i c_j}} - \frac{1 - \mathcal{G}_s^\phi[i, j]}{1 - \Omega_{c_i c_j}} \right] = 0, \quad (19)$$

$$\sum_{\mathcal{G}_s^\phi} q(\mathcal{G}_s^\phi) \sum_{i \neq j} \mathcal{G}_s^\phi[i, j] \left[\frac{N_{ij}}{\alpha} - \frac{N - N_{ij}}{1 - \alpha} \right] = 0, \quad (20)$$

$$\sum_{\mathcal{G}_s^\phi} q(\mathcal{G}_s^\phi) \sum_{i \neq j} (1 - \mathcal{G}_s^\phi[i, j]) \left[\frac{N_{ij}}{\beta} - \frac{N - N_{ij}}{1 - \beta} \right] = 0. \quad (21)$$

By solving these equations, we get:

$$\alpha = \frac{\sum_{i \neq j} \mathcal{A}_k^\phi[i, j] N_{ij}}{N \sum_{i \neq j} \mathcal{A}_k^\phi[i, j]}, \quad (22)$$

$$\beta = \frac{\sum_{i \neq j} (1 - \mathcal{A}_k^\phi[i, j]) N_{ij}}{N \sum_{i \neq j} (1 - \mathcal{A}_k^\phi[i, j])}, \quad (23)$$

$$\Omega_{rs} = \begin{cases} \frac{M_{rs}}{n_r n_s} & \text{if } r \neq s, \\ \frac{2M_{rr}}{n_r(n_r - 1)} & \text{otherwise,} \end{cases} \quad (24)$$

where $n_r = \sum_i \delta_{c_i r}$, $M_{rs} = \sum_{i \neq j} \mathcal{A}_k^\phi[i, j] \delta_{c_i r} \delta_{c_j s}$. Substituting (7), (9), (18) into (12), we obtain the estimated adjacency matrices $\mathcal{A}_s^\phi$ for homogeneous graph $\mathcal{G}_s^\phi$ as follows:

$$\begin{aligned} \mathcal{A}_s^\phi[i, j] &= \\ & \frac{\Omega_{c_i c_j} \alpha^{N_{ij}} (1 - \alpha)^{N - N_{ij}}}{\Omega_{c_i c_j} \alpha^{N_{ij}} (1 - \alpha)^{N - N_{ij}} + (1 - \Omega_{c_i c_j}) \beta^{N_{ij}} (1 - \beta)^{N - N_{ij}}}. \end{aligned} \quad (25)$$

As such, RETA iteratively updates Θ to optimize the graph encoder, generating more accurate observations, and updates $\mathbf{A}$

Algorithm 1: The Workflow of RETA.

Input: Traffic captures $\mathbf{T}$, k in k -NN graph, threshold ϵ , tolerance λ and iterations τ
Initialization: Θ , α , β and Ω
Output: Optimal matrices $\mathcal{A}$ and HAN parameters Θ
Construct $\mathbf{G} = (\mathbf{G}_m, \mathbf{G}_s)$ from $\mathbf{T}$;
Retrieve adjacency matrices $\mathbf{A} = (\mathbf{A}_m, \mathbf{A}_s)$ from $\mathbf{G}$;
 $\mathcal{A} = \mathbf{A}$;
for $i = 1$ **to** τ **do**
 Update Θ via Eq. (14);
 Construct the observation set $\mathbf{O} = \{\mathbf{O}_m, \mathbf{O}_s\}$ via Eq. (4), (5);
 for ϕ in Φ_m **do**
 Update $\mathcal{A}_m^\phi$ by aggregating $\mathbf{O}_m^\phi$ via Eq. (16);
 for ϕ in Φ_s **do**
 while $|\alpha' - \alpha| > \lambda$ **or** $|\beta' - \beta| > \lambda$ **do**
 $\alpha = \alpha'$;
 $\beta = \beta'$;
 Update Ω' , α' and β' via Eq. (22-24);
 Update $\mathcal{A}_s^\phi$ given $\mathbf{O}_s^\phi$ via Eq. (25);
 Sparsification on $\mathcal{A}$ by ϵ via Eq. (13);
return $\mathcal{A}$;

to estimate a better graph topology. The relation subgraphs of heterogeneous and homogeneous graphs are estimated independently. Following an iterative manner, RETA achieves mutual optimizations of both data and model, simultaneously improve relation modeling and detection robustness.

F. Complexity Analysis

The computational complexity of RETA's key components is analyzed as follows. We decompose the overall computational cost into three core stages: graph construction, observation generation, and graph derivation, and provide an in-depth evaluation of the time complexity for each stage. This analysis helps to understand the scalability and efficiency of RETA when applied to large-scale encrypted traffic datasets.

1) *Graph Construction:* Graph construction involves three main steps. First, flow/node features are extracted in linear time with respect to the graph size, i.e., $O(N)$. Second, behavioral similarity between nodes is computed to construct k -NN graphs, which requires $O(kN \log N)$ time due to nearest neighbor search with efficient approximate methods. Third, semantic metapath are constructed with linear complexity $O(N)$. Overall, the dominant term is $O(kN \log N)$ as follows:

$$\begin{aligned} C_{\text{graph}} &= \underbrace{O(N)}_{\text{feature extraction}} + \underbrace{O(kN \log N)}_{\text{behavioral similarity}} + \underbrace{O(N)}_{\text{semantic metapath}} \\ &= O(kN \log N), \end{aligned} \quad (26)$$

where N and E denote the number of nodes and edges, respectively.

2) *Observation Generation*: For a l -layer HAN, each layer involves both node-level aggregation and semantic-level attention across different metapaths. The node-level aggregation requires $O(Nd^2 + Ed)$ per layer due to matrix multiplications and neighbor aggregation. Semantic-level aggregation adds another $O(Nd^2)$. Additionally, during multi-view observation generation, neighbor similarity computations contribute $O(kN \log N)$. However, since d is typically large compared to $k \log N$, the overall complexity is dominated by $O(lNd^2)$ as follows:

$$C_{\text{obs}} = l \cdot \left[\underbrace{O(Nd^2 + Ed)}_{\text{node-level agg.}} + \underbrace{O(Nd^2)}_{\text{semantic-level agg.}} + \underbrace{O(kN \log N)}_{\text{observation generation}} \right] = O(lNd^2), \quad (27)$$

where d denotes node feature dimension.

3) *Graph Derivation*: In the dense case, graph would require $O(N^2)$ per iteration due to pairwise relation updates. To improve scalability, RETA adopts a sparse approximation strategy, where only a limited number of significant neighbors (on average s) are considered per node, reducing the cost to $O(sN)$ per iteration.

$$C_{\text{bayes}} = \tau \cdot \left[\underbrace{O(N^2)}_{\text{dense case}} \rightarrow \underbrace{O(sN)}_{\text{sparse approx.}} \right] = O(\tau sN) \quad (28)$$

where τ is the number of EM iterations, and $s \ll N$ is the average number of neighbors in sparse inference.

V. EXPERIMENTS

A. Dataset and Setups

1) *Dataset*: We evaluate the detection performance and robustness of RETA on three experimental datasets, including two public datasets and one self-collected dataset. The detailed descriptions of these datasets are as follows:

- *DoH* [66] dataset is a DNS over HTTPS (DoH) dataset including benign-DoH, malicious-DoH, and non-DoH traffic generated by five browsers and tools (i.e., Google Chrome, Mozilla Firefox, dns2tcp, dnscat2, and iodine) and four servers (i.e., AdGuard, Cloudflare, Google DNS, and Quad9).
- *MTA* [67] dataset consists of a collection of malware traffic captures published in a blog (<http://malware-traffic-analysis.net>) focusing on network traffic related to malware infections.
- *EC&C* dataset is generated by three customized Remote Administrator Tools (i.e., Metasploit, Pupy, and Quasar-RAT). We randomly generate command sequences, remotely execute them on the compromised hosts, and record the interactive communications.

Table II summarizes the statistical details of the datasets.

2) *SETUPS*: RETA is primarily implemented with PyTorch and Deep Graph Library (DGL). The raw traffic captures are preprocessed and extracted by Tshark. For fair comparisons with

TABLE II
THE STATISTICAL DETAILS OF THE DATASETS

Dataset	#Session		#PCAP
	Benign	Malicious	
<i>DoH</i>	817,300	249,750	31
<i>MTA</i>	$\sim$	132,418	3,117
<i>EC&C</i>	$\sim$	900	3

other baseline models, we utilize PcapPlusPlus, a multiplatform C++ library, to randomly drop packets to simulate network fluctuations, and we utilize DeepRobust, a PyTorch adversarial library for attack and defense methods on graphs, to perform the Fast Gradient Sign Method (FGSM) attack on the graph to simulate adversarial attacks.

Notably, For MTA and EC&C datasets, we collect encrypted traffic captures of various benign applications from the edge routers in Beijing University of Posts and Telecommunications campus network as negative samples.

3) *Baseline Models*: To validate the superior performance of RETA, we compare RETA with five baseline models as follows. Among all baseline models, two are traditional ML-based and DL-based methods that rely on single-session analysis. Another two are GNN-based methods that model multi-session relations as graph structures for correlation analysis. The rest one is a HGNN-based method to validate the robustness of RETA against adversarial attacks on heterogeneous graph. It is notable that most of the selected methods are open-source and we directly utilize their publicly available implementations. For those that are not open-source, we have carefully reproduced them, strictly adhering to the hyper-parameter configurations specified in the original papers. Additionally, all baseline models are fine-tuned to achieve their optimal performance on our experimental datasets. A brief overview of the selected baselines is provided below.

USTC-CNN [28] is an end-to-end encrypted traffic classifier with 1D convolution neural network. Raw traffic captures are processed and reassembled into different sessions. The first 784 bytes in each session are utilized for detection.

IARF [68] is an encrypted malware traffic detection method via Improved Adaptive Random Forests (IARF) to adaptively update parameters when handling unseen malware traffic.

RSGAT [9] is an encrypted malicious traffic detection method based on tailored graph attention networks. RSGAT models the reliability and stationary of traffic communication patterns.

RETRIAL [19] constructs a relational multigraph to model the relations among encrypted flows, and utilize graph structure estimation network for misleading relation correction and selective relation incorporation.

HAN [54] models heterogeneous relations among network objects as metapaths and takes the graph as input and follows a hierarchical attention mechanism to aggregate both metapath-based and similarity-based neighborhood information.

B. Performance Analysis

In this section, we begin by comparing RETA with all baseline models in clean scenario. Then we simulate network fluctuations

TABLE III
PERFORMANCE COMPARISONS OF BASELINE MODELS AND RETA IN DIFFERENT ADVERSE ENVIRONMENTS

DATASET	SCENARIO	USTC-CNN		IARF		RSGAT		ReTRIAL		HAN		RETA	
		ACC	FPR	ACC	FPR	ACC	FPR	ACC	FPR	ACC	FPR	ACC	FPR
<i>DoH</i>	clean	89.43	10.25	98.91	0.59	98.10	1.05	99.57	0.76	99.41	0.63	99.50	0.66
	p=10%	80.58	22.59	97.63	2.63	85.80	1.56	98.06	1.82	96.45	5.35	99.10	0.73
	p=20%	69.66	30.33	95.57	3.04	73.60	13.46	97.59	2.25	94.05	5.57	97.45	1.57
	p=30%	68.08	31.42	94.41	3.41	63.90	39.90	96.81	2.54	90.15	9.25	96.95	1.96
	$\Delta=1$	-	-	-	-	-	-	-	-	94.90	5.55	99.25	0.84
	$\Delta=3$	-	-	-	-	-	-	-	-	93.80	4.65	99.20	0.79
	$\Delta=5$	-	-	-	-	-	-	-	-	92.30	8.05	99.15	0.77
<i>MTA</i>	clean	98.73	0.92	98.94	0.87	99.52	0.57	99.35	0.64	98.93	0.87	99.23	0.69
	p=10%	96.72	5.31	94.27	3.35	98.47	1.07	98.82	0.96	98.41	1.24	99.07	0.68
	p=20%	92.81	7.63	91.72	10.27	97.14	2.37	98.53	1.16	96.92	4.37	98.97	0.74
	p=30%	87.60	17.84	86.42	18.51	95.82	3.17	98.05	1.52	94.51	3.24	98.88	0.87
	$\Delta=1$	-	-	-	-	-	-	-	-	96.27	4.69	99.12	0.71
	$\Delta=3$	-	-	-	-	-	-	-	-	95.81	2.97	99.03	0.56
	$\Delta=5$	-	-	-	-	-	-	-	-	95.15	3.24	98.94	1.29
<i>EC&C</i>	clean	99.76	0.53	99.61	0.57	99.82	0.37	99.86	0.35	99.86	0.41	99.84	0.09
	p=10%	96.37	5.42	95.19	7.53	98.13	0.92	99.21	0.85	98.07	0.94	99.68	0.49
	p=20%	93.48	7.81	94.26	6.84	95.17	6.27	98.57	1.10	96.12	5.76	99.43	0.67
	p=30%	89.45	9.77	90.53	9.62	93.80	8.06	98.23	1.31	93.84	8.39	99.31	0.63
	$\Delta=1$	-	-	-	-	-	-	-	-	97.42	2.99	99.71	0.32
	$\Delta=3$	-	-	-	-	-	-	-	-	96.31	5.81	99.67	0.27
	$\Delta=5$	-	-	-	-	-	-	-	-	95.72	6.53	99.62	0.34

and adversarial attacks to evaluate the performance degradation of RETA and other baseline models. The overall results are summarized in Table III. The specific observations from the experimental results are divided into three parts as follows:

1) *Performance Comparisons in Clean Scenario*: In the absence of adversarial interference, all models achieve strong performance, with graph-based approaches generally outperforming traditional methods. As exhibited in Table III, on the *DoH* dataset, ReTRIAL achieves the highest accuracy (99.57%), followed closely by RETA (99.50%) and HAN (99.41%), while USTC-CNN lags significantly (89.43%). A similar trend is observed on the *MTA* dataset, where RSGAT performs best (99.52%), with RETA (99.23%) and ReTRIAL (99.35%) also showing competitive results. The *EC&C* dataset sees ReTRIAL and HAN tied for the highest accuracy (99.86%), but RETA achieves the lowest false positive rate (0.09%), demonstrating its ability to maintain high precision while minimizing false positives.

These results indicate that graph-based models, particularly those leveraging relational structures (e.g., ReTRIAL, RETA, RSGAT), are highly effective in clean environments. Traditional methods like USTC-CNN struggle due to their limited ability to capture multi-session relations. RETA's strong performance in this scenario validates its design, which integrates semantic metapaths and behavioral similarities without introducing adverse effects. This lays a solid foundation for further robustness evaluation in adverse environments.

2) *Degradations Under Network Fluctuations*: To evaluate the robustness against network fluctuations, we simulate packet loss scenarios with varying dropout rates $p \in \{10\%, 20\%, 30\%\}$. Table III reveals that while all models exhibit performance degradation under these conditions, the extent varies considerably based on both architectural design and dataset characteristics.

DoH dataset, representing DNS-over-HTTPS traffic, proves particularly challenging for most models under packet loss conditions. Here, RSGAT's accuracy plummets to 63.90% at $p = 30\%$, suggesting its graph attention mechanism is highly sensitive to missing packets in encrypted DNS streams. In contrast, RETA maintains 96.95% accuracy, demonstrating exceptional resilience that stems from two key innovations: (1) adaptive neighborhood aggregation that reconstructs missing node features through relational inference, and (2) a dynamic graph completion mechanism that probabilistically estimates disrupted connections, which proves particularly effective for *DoH*, where packet loss often breaks critical temporal dependencies in DNS query sequences.

On *MTA* dataset, we observe a different pattern of degradation. While USTC-CNN and IARF show similar vulnerability (both $< 87\%$ accuracy at $p = 30\%$), the performance gap between RETA (98.88%) and other graph-based models narrows compared to *DoH*. This suggests that malware traffic patterns in *MTA* may contain more redundant behavioral signatures, making them inherently more tolerant to packet loss. Nevertheless, RETA's consistent advantage highlights its ability to leverage both structural and temporal regularities in malicious traffic sessions.

EC&C dataset presents the most stable performance landscape across models, with even baseline methods maintaining $> 90\%$ accuracy at maximum packet loss. However, RETA still achieves superior results (99.31%), likely because covert-channel communications exhibit more predictable periodicity in their communication patterns. The model's channel attention mechanism appears particularly effective at reconstructing irregular transmission patterns when packets are dropped.

Notably, the comparison with HAN reveals fundamental architectural advantages. Where HAN's static graph construction fails to adapt to missing data (93.84% on *EC&C* at $p = 30\%$),

RETA's probabilistic edge estimation maintains graph integrity through: (1) learned attention weights that redistribute importance across surviving connections, and (2) meta-path guided feature imputation that preserves semantic relations. This explains the $<3\%$ accuracy drop for RETA across all datasets at maximum packet loss, compared to 5-9% drops for other models.

These results collectively demonstrate that while dataset-specific traffic patterns influence absolute performance, RETA's design principles provide consistent advantages across diverse encrypted communication scenarios. The framework's ability to dynamically reconfigure its graph representation in response to missing data makes it particularly valuable for real-world deployment, where network conditions are unpredictable and packet loss is inevitable.

3) Degradations Under Adversarial Attacks: Following [12], we employ FGSM to generate adversarial perturbations with $\Delta \in \{1, 3, 5\}$ edge modifications. To ensure fair comparison across models with different graph construction methodologies, we focus our adversarial evaluation on RETA and its closest counterpart HAN. The results reveal striking differences in adversarial robustness that vary meaningfully across datasets.

On the *DoH* dataset, which exhibits complex query-response dependencies, RETA maintains 99.15% accuracy even at $\Delta = 5$, demonstrating remarkable robustness against edge perturbations. In contrast, HAN's accuracy degrades to 92.30% under identical conditions. This substantial 6.85% performance gap highlights how RETA's graph structure optimization preserves detection performance when adversarial edges disrupt normal DoH traffic patterns.

The *EC&C* dataset shows a similar trend, with RETA (99.62%) outperforming HAN (95.72%) by 3.9%. This smaller margin suggests that covert-channel communications may be inherently more robust to edge perturbations, possibly due to more standardized protocol structures. Nevertheless, RETA's dynamic structure learning mechanism proves consistently advantageous, automatically identifying and downweighting suspicious edges through its attention-based filtering.

Interestingly, the malware-focused *MTA* dataset reveals the narrowest performance gap (98.94% vs. 95.15% at $\Delta = 5$). This relative similarity implies that malicious traffic patterns in *MTA* may contain more distinctive behavioral signatures that remain detectable even when edges are perturbed. However, RETA still maintains its advantage by combining three key defensive mechanisms: (1) multi-view relation modeling including semantic metapaths and behavioral similarities, (2) mutual optimization of both graph data and detection model.

These dataset-specific variations underscore that while adversarial vulnerability depends partly on traffic features, RETA's architectural innovations provide consistent protection across different encrypted communication scenarios. The results particularly emphasize how static graph approaches like HAN struggle with edge perturbations, while RETA's dynamic graph structure learning offers crucial adaptability against evolving evasion techniques.

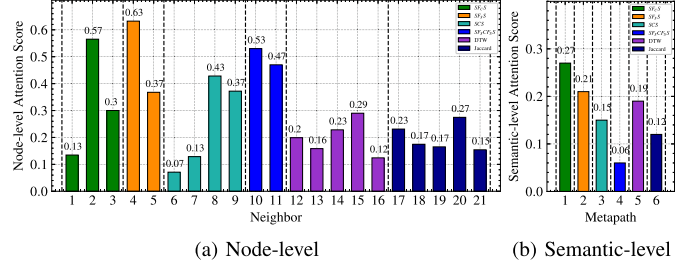


Fig. 4. Visualization of attention scores.

C. Interpretability Analysis

To improve the interpretability of RETA in encrypted malicious traffic detection, we incorporate a detailed analysis of its hierarchical attention mechanism, including both node-level and semantic-level attention. This analysis aims to visualize how RETA prioritizes neighbors and relations among encrypted sessions during the detection process, and to provide insights into which factors contribute most to decision-making, improving transparency and trustworthiness in its predictions.

Node-level Attention: RETA captures discriminative local structures by assigning attention to neighbors under each metapath, encompassing both semantic paths and behavioral similarity relations. Since each metapath induces a distinct subgraph, the corresponding neighborhoods differ. As illustrated in Fig. 4(a), attention scores are computed independently per metapath. Notably, metapath-based neighbors exhibit varied contributions, reflecting diverse structural semantics, whereas similarity-based neighbors tend to receive more uniform attention due to their less discriminative nature. The attention mechanism enables RETA to distinguish informative neighbors from noisy ones, especially under complex contexts.

Semantic-level Attention: While node-level attention captures fine-grained local patterns, semantic-level attention aggregates the meta-path-specific embeddings to learn the relative importance of different semantic contexts.

As illustrated in Fig. 4(b), we observe that *SFCS* meta-path receives the highest attention score among all relations, suggesting that client-side TLS fingerprints are the most discriminative feature in distinguishing malware-generated traffic from benign traffic. The *SFSS* and *SCS* paths contribute moderately, indicating server configuration and certificate identity still provide valuable but less dominant signals. In contrast, the transitive meta-path *SFSCFS* exhibits a relatively low attention score. A possible reason is that the occurrence of multi-hop associations is infrequent in real-world traffic.

Additionally, the behavioral similarity relation based on DTW distance achieves a moderate attention score, implying that temporal similarity in packet length sequences captures critical patterns such as consistent upload/download behaviors in coordinated attacks. While the relation based on Jaccard distance is assigned a slightly lower attention score, which may be attributed to its limited expressiveness in capturing nuanced variations under adversarial conditions.

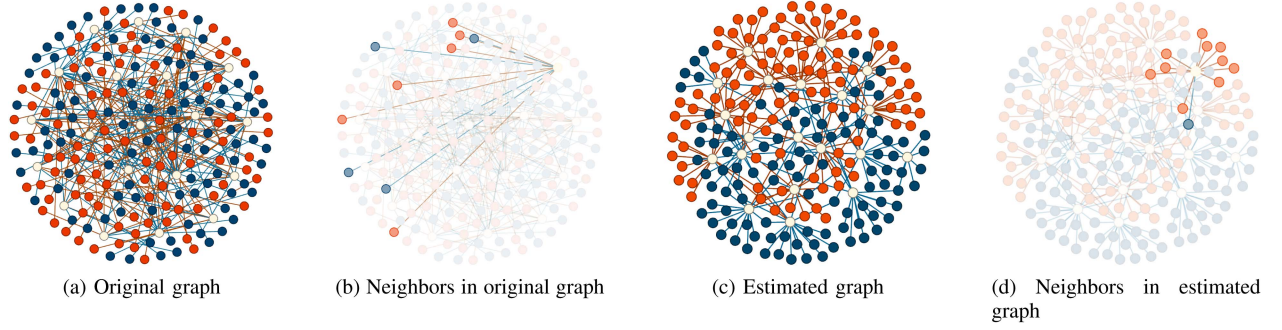


Fig. 5. Visualization of the graph structures for (a) original heterogeneous subgraph of metapath SF_sS and (c) corresponding estimated graph generated by RETA, where red indicates the benign session, blue indicates the malicious session, beige indicates the server fingerprint. The remaining nodes and edges are faded to highlight the heterogeneous neighbors of selected server fingerprint node in (b) original graph and (d) estimated graph.

TABLE IV
PERFORMANCE COMPARISONS OF RETA VARIANTS IN DIFFERENT ADVERSE ENVIRONMENTS

DATASET	SCENARIO	w/o M		w/o S		RETA	
		ACC	FPR	ACC	FPR	ACC	FPR
DoH	clean	99.30	0.69	99.05	0.75	99.50	0.66
	p=10%	97.30	1.38	95.15	3.16	99.10	0.73
	p=20%	95.70	2.82	94.18	1.70	97.45	1.57
	p=30%	94.25	3.22	94.15	1.75	96.95	1.96
	$\Delta=1$	99.20	0.79	98.50	1.14	99.25	0.84
	$\Delta=3$	98.80	1.27	98.01	2.28	99.20	0.79
	$\Delta=5$	98.70	1.18	97.56	2.41	99.15	0.77
MTA	clean	98.87	0.84	99.01	0.74	99.23	0.69
	p=10%	98.67	1.05	98.33	0.92	99.07	0.68
	p=20%	98.12	0.97	97.82	1.29	98.97	0.74
	p=30%	97.41	2.26	95.14	3.48	98.88	0.87
	$\Delta=1$	98.85	0.84	98.76	1.25	99.12	0.71
	$\Delta=3$	98.47	1.53	98.42	1.48	99.03	0.56
	$\Delta=5$	98.33	0.97	98.40	1.49	98.94	1.29
EC&C	clean	99.89	0.24	99.96	0.12	99.84	0.09
	p=10%	98.42	0.83	99.52	0.64	99.68	0.49
	p=20%	95.63	6.34	99.23	0.71	99.43	0.67
	p=30%	94.97	7.36	98.52	0.93	99.31	0.63
	$\Delta=1$	99.41	0.52	99.57	0.45	99.71	0.32
	$\Delta=3$	99.23	0.56	99.48	0.63	99.67	0.27
	$\Delta=5$	99.21	0.71	99.42	0.67	99.62	0.34

D. Ablation Study

We conduct an ablation study to evaluate the contribution of each key component in RETA. Specifically, we first compare the full RETA with two variants as follows:

- w/o M removes homogeneous optimization (i.e., Bayesian inference) on similarity-based relation subgraphs.
- w/o S removes heterogeneous optimization (i.e., channel attention) on metapath-based relation subgraphs.

As summarized in Table IV, the full RETA model exhibits minimal performance degradation under adversarial conditions. With a 30% packet loss rate, its accuracy drops by only 1.14% on average across the three datasets, demonstrating strong robustness. In contrast, the variants show significantly larger declines. For example, on EC&C dataset, w/o M suffers a 4.92% accuracy drop (from 99.89% to 94.97%), while w/o S degrades by 1.44% (from 99.96% to 98.52%). This highlights the complementary roles of both optimizations in maintaining robustness.

TABLE V
THE OPTIMAL HYPER-PARAMETERS PER DATASET

Dataset	k	ϵ	λ	ACC
DoH	6	0.8	10^{-2}	99.50
MTA	6	0.7	10^{-3}	99.23
EC&C	7	0.8	10^{-4}	99.65

k : Large k suits dense graphs; small k avoids noise in sparse graphs.
 ϵ balances edge retention and sparsity.
 λ balances precision and computational cost.

E. Graph Visualization

We take MTA dataset as an example and utilize Gephi to visualize SF_sS metapath-based heterogeneous subgraph and DTW distance-based homogeneous subgraph to show the graph structure optimizations by RETA. As shown in Fig. 5(a) and (c), the original graph is chaotic, while the estimated graph is more harmonious. Zooming into the neighbors of the selected server fingerprint node, as shown in Fig. 5(b) and (d), the server fingerprint node connects with both benign and malicious session nodes in the original graph. In contrast, the metapath-based neighbors tend to have the same labels in the estimated graph. Likewise, as shown in Fig. 6, with the optimization of RETA, homogeneous nodes with same labels are more likely to be connected. The estimated homogeneous graph is more in line with the well-performed premise of HGNNs.

F. Hyper-Parameter Sensitivity

In this subsection, we present the rigorous hyper-parameter tuning process in RETA using grid search with 5-fold cross-validation on each dataset's training split. The hyper-parameters include $k \in [5, 6, \dots, 9]$ for k -NN graph construction, $\epsilon \in [0.6, 0.7, \dots, 0.9]$ for graph sparsification, and $\lambda \in [10^{-5}, \dots, 10^{-1}]$ for convergence control. Table V summarizes the optimal values per dataset and selection rationale.

k : As illustrated in Fig. 7(a), generally, with the increment of k value, the accuracies of RETA on three datasets show an ascending trend at the beginning, then gradually decrease. RETA achieves the best performance with $k = [6, 6, 7]$ on DoH, MTA, and EC&C datasets. We believe that the constructed

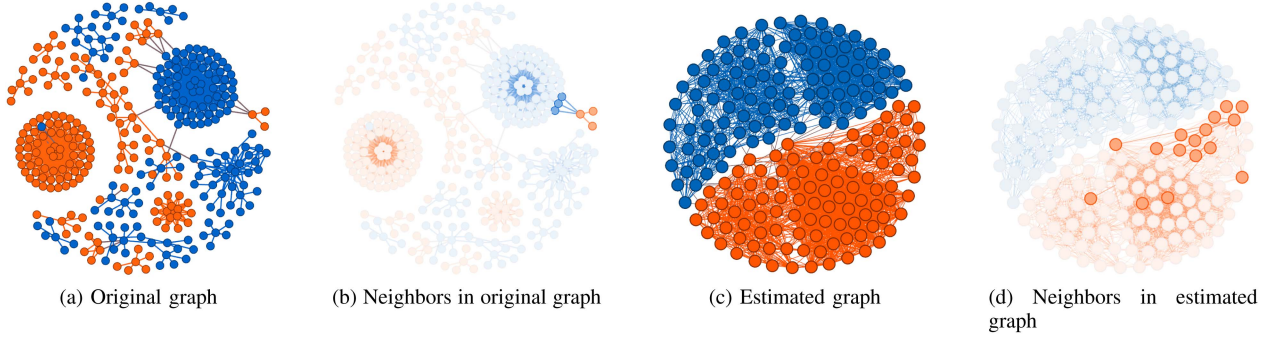


Fig. 6. Visualization of the graph structures for (a) original homogeneous subgraph with DTW distance and (c) corresponding estimated graph generated by RETA, where red indicates benign sessions, blue indicates malicious sessions. The remaining nodes and edges are faded to highlight the homogeneous neighbors of selected benign session node in (b) original graph and (d) estimated graph.

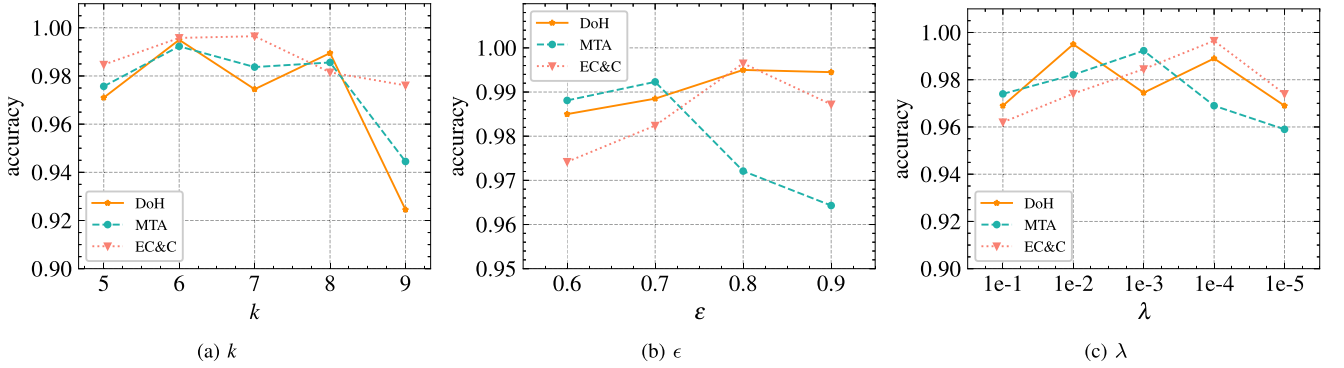


Fig. 7. Hyper-parameters sensitivity analysis on *DoH*, *MTA*, and *EC&C* datasets.

k -NN graphs with larger k value incorporate more structural information within multi-order neighborhoods, thus achieving better detection performance. However, excessive correlations may introduce more noise and mislead the decisions of HGNN.

ϵ : Additionally, we can see that the sparsification threshold ϵ also dramatically affects the RETA performance in Fig. 7(b). RETA achieves the best performance with $\epsilon = [0.8, 0.7, 0.8]$ on *DoH*, *MTA*, and *EC&C* datasets, respectively. Similarly, the accuracies of RETA first increase then decrease because too sparse or too dense graph structure with too little or too large ϵ would lead to unsatisfactory performance.

λ : λ controls the convergence of RETA. When the difference between the updated and the original parameters is less than λ , the optimization of RETA ends. As shown in Fig. 7(c), the optimal ϵ values are $[10^{-2}, 10^{-3}, 10^{-4}]$ respectively on *DoH*, *MTA* and *EC&C* datasets.

G. Embedding Visualization

To intuitively evaluate the effectiveness of our proposed RETA, we visualize the embeddings of encrypted sessions learned by RETA, $RETA_m$, $RETA_s$, HAN in adverse environments by t-SNE [69]. Specifically, we take *MTA* dataset as an example to present the representations of RETA and other baseline models in the case that the packet loss rate is 10%. As illustrated in Fig. 8, the orange points represent the embeddings of encrypted benign sessions, and the green points represent the embeddings of encrypted malicious sessions.

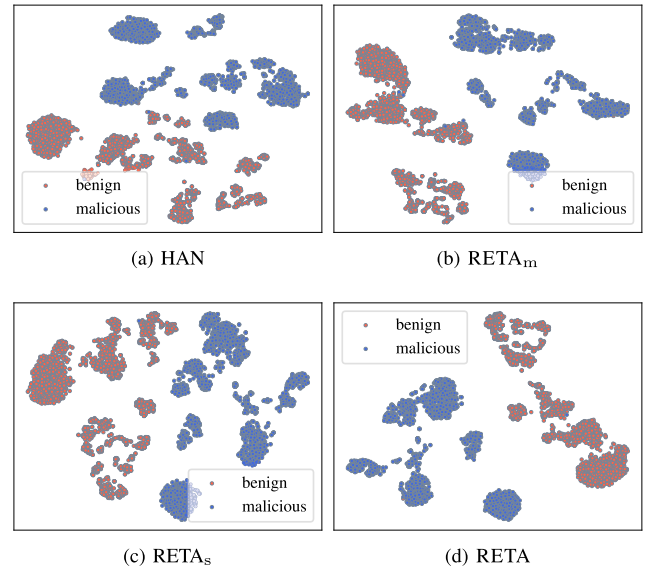


Fig. 8. Embedding visualization of HAN, $RETA_m$, $RETA_s$ and RETA.

We can see that packet loss causes the loss of valuable information and affect the decisions of detectors to a great extent. However, our proposed RETA effectively tackles this adverse environment. The visualization of the embeddings learned by RETA has the most apparent boundaries between benign and malicious.

VI. DISCUSSION

RETA exhibits superior robustness in encrypted malicious traffic detection under adversarial network conditions, yet it still exposes some potential issues to be addressed. In this section, we provide a critical analysis of RETA's limitations and present promising directions for future research.

1) *Graph Paradigm*: RETA constructs a comprehensive graph that integrates both heterogeneous and homogeneous subgraphs to capture semantic metapaths and behavioral similarities among encrypted sessions. The former are derived from SSL/TLS protocol-related fields, while the latter are measured by packet length sequences with various distance metrics. However, as a general detection system, RETA should not be constrained to specific protocols or predefined relations. Future research could explore protocol-independent metapaths and additional distance metrics to model richer relations among encrypted sessions for subsequent analysis.

2) *Unified Optimization*: By employing a unified graph structure learning approach, RETA decomposes the optimization of the graph into independent sub-optimizations for its heterogeneous and homogeneous subgraphs divided by relations. Specifically, heterogeneous subgraphs are estimated using channel attention, while homogeneous subgraphs are derived independently through Bayesian inference independently. Future work should focus on developing a more unified graph optimization framework, incorporating attention-based aggregation and probability-based estimation, to facilitate the joint optimization of relation subgraphs.

3) *Scalability Concerns*: Though RETA achieves strong detection robustness, scalability issues persist when extending to large graphs including the significant computational overhead associated with k -NN calculations during homogeneous subgraph construction and observation generation, and the substantial memory requirements imposed by GNNs. We experiment with a flow aggregation technique that explicitly encodes flow count information into node embeddings, effectively reducing graph scale with slight performance impact. Future research should explore alternative flow aggregation strategies and novel graph construction paradigms to enhance scalability without compromising detection efficacy.

4) *Attack Simulations*: To evaluate RETA's robustness, we simulate network fluctuations and adversarial attacks by dropping packets and adding perturbation edges. Future work should expand these simulations to encompass more realistic attack scenarios, such as DDoS and botnet attacks. Additionally, topological adversarial attacks that mimic legitimate service interactions require further investigation. A broader range of attack simulations will facilitate a more comprehensive assessment of RETA's robustness and contribute to the development of a more practical and robust encrypted malicious traffic detection system to tackle adversarial environments. b

VII. CONCLUSION

In this paper, we propose RETA, a unified framework for robust encrypted malicious traffic detection in adverse environments. Specifically, we construct a comprehensive graph integrating both heterogeneous and homogeneous subgraphs to

model the rich semantic metapaths and behavioral similarities among encrypted sessions. Then we utilize a tailored HAN as graph encoder and employ channel attention-based aggregation and Bayesian inference-based estimation to jointly correct noisy relations. Following an iterative manner, the graph data and detection model achieve mutual optimizations with noisy relation correction and detection robustness improvement. In evaluations, we modify raw traffic captures to simulate network fluctuations and adversarial attacks. Extensive experiments demonstrate the superior robustness of RETA in adverse environments, underscore the potential of RETA to advance encrypted traffic analysis in increasingly adversarial network landscapes. Future work will explore the extension of RETA to more dynamic and complex threat environments, further enhancing its efficiency, adaptability, and resilience.

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